

It's easy to destroy transistors by connecting them incorrectly, so it's important to be careful. In addition, integrated circuits (ICs) can get damaged by ESD when handled by hand. It is therefore a good practice to make sure that the components that you plan to work with are functioning properly before you construct your circuit. This may save you a lot of time and effort in debugging your circuit later on.

Diode: a multimeter or ammeter

1. Identify the Anode and Cathode:

Diodes have two leads, the anode (positive) and the cathode (negative). Identify these leads on the diode. The cathode end is usually marked with a band around it or a different color compared to the anode end.

2. Test Forward Bias:

Place the red probe (positive) of your multimeter on the anode of the diode and the black probe (negative) on the cathode. Your multimeter will display a voltage drop if the diode is functioning correctly. Typically, this voltage drop should be around 0.6 to 0.7 volts for a standard silicon diode.

3. Test Reverse Bias:

Reverse the probes. Place the red probe on the cathode and the black probe on the anode. In this configuration, a healthy diode should display an "overload" on the multimeter, indicating a high resistance, meaning it's not conducting in the reverse direction.

MOSFET: a multimeter or ammeter

1. Identify the gate, drain and source terminal of the device under test.

2. Turn the Multimeter to 'diode' mode.

3. Then, connect the red lead of your digital multimeter with the drain terminal and the black lead with the source terminal. If there is no continuity between these two terminal, then the MOSFET is fine, since there is no gate-source voltage you have applied to turn on the MOSFET. Otherwise if some continuity is shown by the multimeter with some finite value, it means your device is defective.

4. Using an ammeter verify that there is close to infinite resistance between the gate to source and gate to drain. If not, the device is faulty.

OP27: an ammeter

1. Identify the correct pins from the data sheet.
2. Verify that there is no short circuit between the positive supply, negative supply and ground terminals (very high resistance).
3. Verify that there is no short circuit between the positive input, negative input and ground terminals (very high resistance).
4. If in any case you found a short-circuit, it means the IC is faulty.

BJT: a multimeter or ammeter

1. Resistance: For a PNP transistor, the resistance should be low when the negative probe is on the base and the positive probe is on the emitter or collector. If the resistance is high or infinite, the BJT may be damaged. For an NPN the polarities are switched.
2. Voltage drop: Measure the voltage drop between the emitter/source and base pins, and also between the collector/drain and base. A good transistor should have a low voltage drop in one direction and a higher voltage in the reverse direction (according to the PN junction polarities for either NPN or PNP).
3. Diode test: Switch the multimeter to the appropriate diode test mode, connect the red diode to the base, and the black one to the emitter. The meter should show a specific DC voltage value (about 0.7V) that should be compared with the data in the transistor's technical documentation.